Nptel Online Certification Course
Indian Institute of Technology Kharagpur
Computer Vision
Assignment - Week 8

Number of questions: 10 **Total marks: 10x2=20**

FOR QUESTIONS 1 AND 2:

Consider a pixel p in RGB space given by $\begin{bmatrix}40\\50\\30\end{bmatrix}$. Based on the given data solve the following questions 1 and 2:

QUESTION 1: **Type:Comprehensive**

What is the saturation value of this pixel in HSV color space?

a) 0.15

b) 0.25

c) 0.33

d) 0.45

Correct Answer: b)
Detailed Solution:
$V = \frac{R+G+B}{3} = 40.$
$S = 1 - \frac{min(R,G,B)}{V} = 0.25.$

QUESTION 2: **Type:Comprehensive**

What is the corresponding value of the pixel p in CMY space?

a) $\begin{bmatrix}215\\105\\165\end{bmatrix}$

b) $\begin{bmatrix}200\\150\\225\end{bmatrix}$

c) $\begin{bmatrix}115\\205\\165\end{bmatrix}$

d) $\begin{bmatrix}215\\205\\225\end{bmatrix}$

Correct Answer: d)

Detailed Solution:

$$\begin{bmatrix}C\\M\\Y\end{bmatrix} = \begin{bmatrix}255\\255\\255\end{bmatrix} - \begin{bmatrix}R\\G\\B\end{bmatrix} = \begin{bmatrix}215\\205\\225\end{bmatrix}.$$

QUESTION 3: **Type: MSQ**

Which of the following statements are correct?

a) Color captured in the RGB color space is not suitable for direct interpretation of color components.

b) CIE Chromaticity Chart represents colors in a 2-D space according to tri-stimulus model of color representation.

c) In digital cameras color images are mostly captured using a CFA.

d) Luminance Y is encoded using less bits than chrominance values I and Q.

Correct Answer: a), b), c)

Detailed Solution:

Luminance Y is encoded using more bits than chrominance values I and Q.

QUESTION 4: **Type: Numeric**

Interpolate G_5 using laplacian corrected edge correlated interpolation using given Bayer's pattern.

$$\text{Bayer's Pattern} = \begin{bmatrix} & & B_1 & & \\ & & G_2 & & \\ B_3 & G_4 & B_5 & G_6 & B_7 \\ & & G_8 & & \\ & & B_9 & & \end{bmatrix}$$

$$\text{Corresponding CFA values} = \begin{bmatrix} & & 32 & & \\ & & 90 & & \\ 32 & 72 & 90 & 130 & 136 \\ & & 94 & & \\ & & 26 & & \end{bmatrix}$$

Correct Answer: 104

Detailed Solution:

$\Delta H = |G_4 - G_6| + |B_5 - B_3 + B_5 - B_7| = 70$

$\Delta V = |G_2 - G_8| + |B_5 - B_1 + B_5 - B_9| = 126$

As $\Delta H < \Delta V$

$G_5 = \frac{G_4+G_6}{2} + \frac{B_5-B_3+B_5-B_7}{4} = 104$

QUESTION 5: **Type: MCQ**

Consider that the color of a source illuminant is represented by an RGB vector $(120, 160, 200)$, whereas the target color is given by $(60, 120, 150)$. Given a color vector $(120, 80, 160)$ in the source image, compute the color corrected vector using diagonal correction rule.

a) $(90, 90, 180)$

b) $(75, 150, 75)$

c) $(50, 100, 50)$

d) $(100, 50, 100)$

Correct Answer: a)

Detailed Solution:

$k_r = \frac{R_d}{R_s} = 0.5, \quad k_g = \frac{G_d}{G_s} = 0.75, \quad k_b = \frac{B_d}{B_s} = 0.75$

where, R_d represent the R value of the target color vector and R_s represent the R value of the source color vector.

$f = \frac{R+G+B}{k_r R + k_g G + k_b B} = 1.5$

$R_u = f k_r R, \quad G_u = f k_g G, \quad B_u = f k_b B$

QUESTIONS 6: **Type: MSQ**

Which of the following statements are true?

a) Image reconstruction using color demosaicing leads to sharp edges in an image.

b) Green channel is less likely to be aliased.

c) In CFA, green channels are sampled at a higher rate than red and blue.

d) High correlation between red, green and blue channels are very likely to have similar texture and edge locations.

Correct Answer: b), c), d)

Detailed Solution:

Image reconstruction using color demosaicing lead to blurred edges in an image. In CFA, green channels are sampled at a higher rate than red and blue. High correlation between red, green and blue channels are very likely to have similar texture and edge locations. Green channel is less likely to be aliased and details are preserved better in green channels compared to red or blue.

FOR QUESTIONS 7 AND 8:

Given Bayer's pattern.

$$\text{Bayer's Pattern} = \begin{bmatrix} G_1 & R_2 & G_3 & R_4 & G_5 \\ B_6 & G_7 & B_8 & G_9 & B_{10} \\ G_{11} & R_{12} & G_{13} & R_{14} & G_{15} \\ B_{16} & G_{17} & B_{18} & G_{19} & B_{20} \\ G_{21} & R_{22} & G_{23} & R_{24} & G_{25} \end{bmatrix}$$

$$\text{Corresponding CFA values} = \begin{bmatrix} 10 & 14 & 18 & 24 & 30 \\ 15 & 25 & 35 & 45 & 55 \\ 24 & 45 & 150 & 225 & 168 \\ 65 & 87 & 124 & 100 & 144 \\ 34 & 225 & 200 & 235 & 245 \end{bmatrix}.$$

Based on the given information, solve questions 7 and 8.

QUESTION 7: **Type: Comprehensive**

Using interpolation by averaging red and blue hues, compute G_8.

Correct Answer: 59.5

Detailed Solution:

$G_8 = \frac{G_3+G_7+G_9+G_{13}}{4}$.

QUESTION 8: **Type: Comprehensive**

Using interpolation by averaging red and blue hues, compute B_{12}.

Correct Answer: 64.7

Detailed Solution:

$B_{12} = \frac{G_{12}}{4}(\frac{B_6}{G_6} + \frac{B_8}{G_8} + \frac{B_{16}}{G_{16}} + \frac{B_{18}}{G_{18}})$.

QUESTION 9: **Type: MCQ**

Given the coordinates in the normalized x-y chromaticity space of three primary colors as $(2/3, 1/3)$, $(1/5, 3/4)$, and $(1/6, 1/10)$, compute the corresponding maximally saturated color in the x-y chromaticity space preserving the same hue and intensity for the given point $(0.31, 0.27)$ in x-y chromaticity space. Evaluate all the results upto three decimal places.

a) approximately $(0.489, 0.308)$

b) approximately $(0.708, 0.186)$

c) approximately $(0.269, 0.154)$

d) approximately $(0.208, 0.286)$

Correct Answer: c)

Detailed Solution: Given the three primary colors as $R = (2/3, 1/3)$, $G = (1/5, 3/4)$, and $B = (1/6, 1/10)$ in x-y chromaticity space. We have white point $w = (1/3, 1/3)$ and a given point $q = (0.31, 0.27)$. If you move the point radially to the gamut edge, you will find a Maximum Saturation given a hue. Intersection of point between the line formed by an edge of the triangle and wq, will be the required answer.

Use the projective space concept of cross product between two points will give you a line passing through the points.

$BG = (1/6, 1/10, 1) \times (1/5, 3/4, 1) = (-0.65, 0.033, 0.105)$

$BR = (-0.233, 0.5, -0.011)$

$GR = (0.417, 0.467, -0.433)$

$wq = (0.063, -0.023, -0.013)$

$BG \times wq = (0.002, -0.002, 0.013) \rightarrow$ Intersection at $(0.154, -0.154)$.

$GR \times wq = (-0.016, -0.022, -0.039) \rightarrow$ Intersection at $(0.410, 0.564)$.

$BR \times wq = (-0.007, -0.004, -0.026) \rightarrow$ Intersection at $(0.269, 0.154)$.

QUESTION 10: **Type:Numeric**

Given Bayer's pattern.

$$\text{Bayer's Pattern} = \begin{bmatrix} G_1 & R_2 & G_3 & R_4 & G_5 \\ B_6 & G_7 & B_8 & G_9 & B_10 \\ G_{11} & R_{12} & G_{13} & R_{14} & G_{15} \\ B_{16} & G_{17} & B_{18} & G_{19} & B_{20} \\ G_{21} & R_{22} & G_{23} & R_{24} & G_{25} \end{bmatrix}$$

$$\text{Corresponding CFA values} = \begin{bmatrix} 10 & 20 & 18 & 24 & 30 \\ 15 & 25 & 35 & 45 & 55 \\ 24 & 45 & 150 & 200 & 168 \\ 65 & 87 & 124 & 100 & 144 \\ 34 & 225 & 200 & 235 & 245 \end{bmatrix}.$$

Interpolate R_{19} and R_7 using billinear interpolation, and compute value of R_{19}+R_7 (till 1 decimal place).

Correct Answer: 250.0

Detailed Solution:

$R_{19} = \frac{R_{14}+R_{24}}{2}$. $R_7 = \frac{R_2+R_{12}}{2}$.
